1 A battery provides energy to an electric car.

(a) The electric car has an acceleration of $2.9\,m/s^2$ when it moves from rest. The combined mass of the car and its driver is 1600 kg.

(i) Calculate the time taken to reach a speed of 28 m/s.

time = ... [2]

(ii) Calculate the force required to produce this acceleration.

force = ... [2]

(iii) Calculate the kinetic energy of the car when its speed is 28 m/s.

kinetic energy = ... [2]

(b) The time taken for the car battery to be recharged from zero charge to full charge is 8.3 h. The charge is delivered to the battery by a charger with a current of 32 A.

Calculate the charge supplied by the charger.

charge = ... [3]

(c) Under ideal conditions, the car can travel a maximum distance of 390 km when the battery is fully charged.

Suggest why, in normal use, the car needs to be recharged after travelling less than 390 km.

.....

..... [1]

[Total: 10]

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